

Fine Grained Concept Learning in Videos

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Abstract

The personalization of generative models has enabled fine-grained control over image and video synthesis. While existing approaches are effective at learning entire objects from videos or localized concepts from single images, learning fine-grained visual concepts directly from video data remains challenging. In this project, we study the problem of extracting a localized visual concept—such as a logo, texture, or decorative pattern—from a single source video and transferring it to novel target videos.

We build our approach on top of the VACE[4] framework and explore weight-space concept learning by finetuning the VACE video control unit using LoRA[3], while keeping the base video diffusion model frozen. To better focus the learning on concept-relevant regions, we adopt a dual masked MSE loss that emphasizes reconstruction within the masked area and reduces the influence of background regions. This setup allows the model to aggregate appearance information across frames and learn a concept that is more consistent under changes in pose and motion.

We evaluate our method on several concept types with varying complexity, including simple patterns, complex logos, abstract structural designs, and dynamic objects. Our experiments show that the proposed approach is effective for transferring high-level visual style and structural characteristics across videos, while exact replication of fine-grained details remains difficult. These results highlight both the strengths and limitations of lightweight LoRA-based adaptation for fine-grained video concept learning. We release our project as open source to facilitate reproducibility and future research, available at <https://github.com/cs-mshah/DiffSynth-Studio/tree/vlr-proj>.

1. Related Works

Recent advances in diffusion-based generation have enabled fine-grained personalization and local editing in images and videos. Methods such as Textual Inversion [2] and DreamBooth [9] introduce text tokens or fine-tuned layers to represent user-specific concepts, while Custom Dif-

fusion [5] and LoRA [3] improve efficiency by optimizing only partial network parameters.

Building upon this line of work, CLiC (Concept Learning in Context) [10] proposes learning localized visual concepts directly within their object context, allowing consistent transfer of local patterns across different objects, such as textures or ornaments. Dynamic Concepts [1] extends this idea to the temporal domain by introducing a two-stage *Set-and-Sequence* framework. It learns dynamic concepts, which capture both the appearance and motion of a subject, from a single input video, allowing spatio-temporally coherent personalization. However, the method focuses on modeling concept-specific dynamics within a fixed video context rather than providing a unified interface for general video editing or composition.

In contrast, unified frameworks like VACE [4] and MagicEdit [6] extend diffusion transformers to handle diverse video generation and editing tasks. These models achieve strong temporal consistency and support multiple conditioning modalities (e.g. depth, pose, layout), but they primarily address global appearance or structure rather than localized concept transfer or dynamic concept reuse. Our project aims to bridge this gap by adapting the VACE control unit to learn a personalized fine grained concept and transfer it across diverse video contexts.

2. Methodology

As shown in figure 1, given a source video, we wish to learn a concept such as a design through an input user mask across the frames (obtained from SAM 2 [8] or any other methods). Once this concept is learned, we wish to edit another video with this learned concept. We are inspired from CLiC [10] and wish to extend its logic for in-context concept learning from images, to videos.

2.1. Datasets

Our fine-grained concept learning framework is trained on a triplet of inputs:

1. a ground truth video with desired style,
2. a binary mask indicating the localized concept to be learned, and

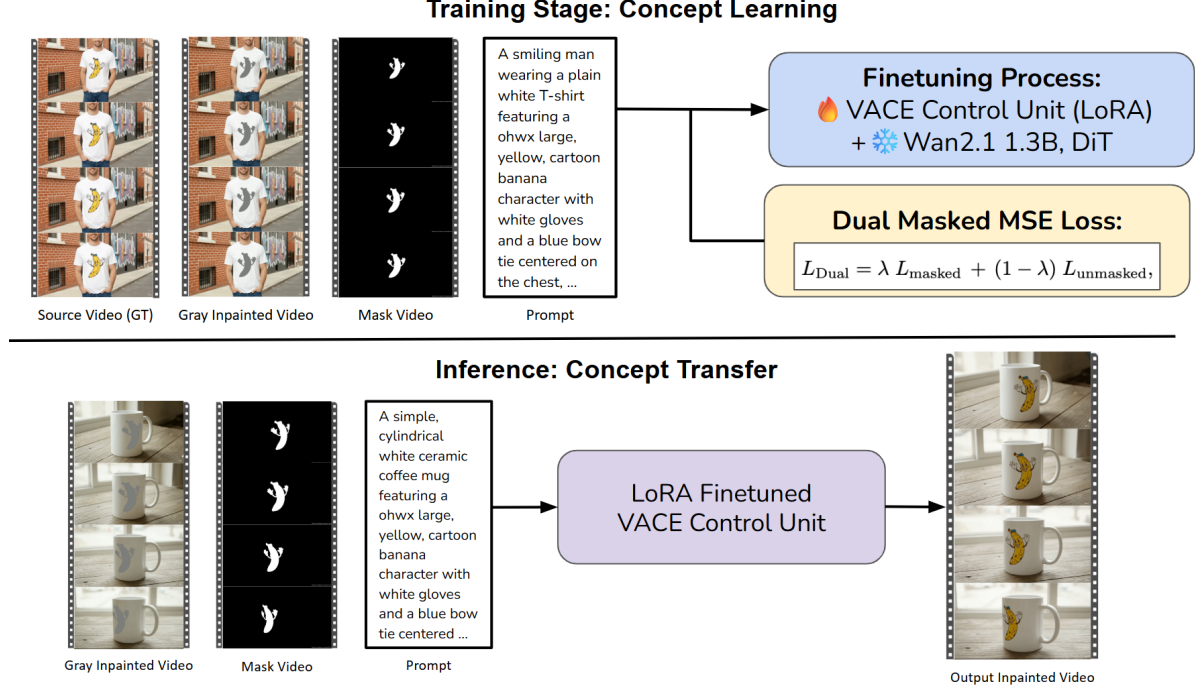


Figure 1. Overview of our training and inference pipeline for fine-grained video concept learning.

3. a gray inpainted conditioning video where the masked region is replaced with a neutral gray value.

For effective and high-quality training, we engineered two distinct pipelines to augment the source videos. While both approaches were explored, we observed that the second method (Generative Augmentation) significantly outperformed the first (Compositing-Based) in terms of realism and training stability.

2.1.1 Method 1: Compositing-Based Augmentation

In the first approach, we construct training samples by compositing segmented object videos onto diverse background videos using affine transformations. Given an object video and its binary mask (obtained via SAM 2 [8]), we apply scale and translation perturbations and overlay the result onto background videos. The gray inpainted conditioning video is produced by replacing the masked region with a constant gray value.

However, as shown in Fig. 2, this compositing-based strategy often produces visually implausible artifacts and inconsistent lighting, which negatively impacts training stability.

2.1.2 Method 2: Generative Augmentation

To overcome the limitations of compositing, we propose a fully generative data pipeline, illustrated in Fig. 3. This



Figure 2. **Failure cases of Method 1.** Each row corresponds to a video

pipeline synthesizes both source videos and target concepts using off-the-shelf image and video generation models.

First, we generate concept images from text prompts using Nano Banana Pro, ensuring multi-view coverage of the target concept. These images are then converted into temporally coherent source videos using Google Veo 3.1. Next, we apply SAM 2 [8] to obtain binary masks that precisely localize the target concept across frames. Finally, the gray inpainted conditioning video is constructed by replacing the masked region with a neutral gray value, following the VACE input format [4].

All generated videos, masks, conditioning inputs, and corresponding text prompts are indexed in a CSV file for training. The full dataset is publicly available at: [Hugging-Face Dataset](#)

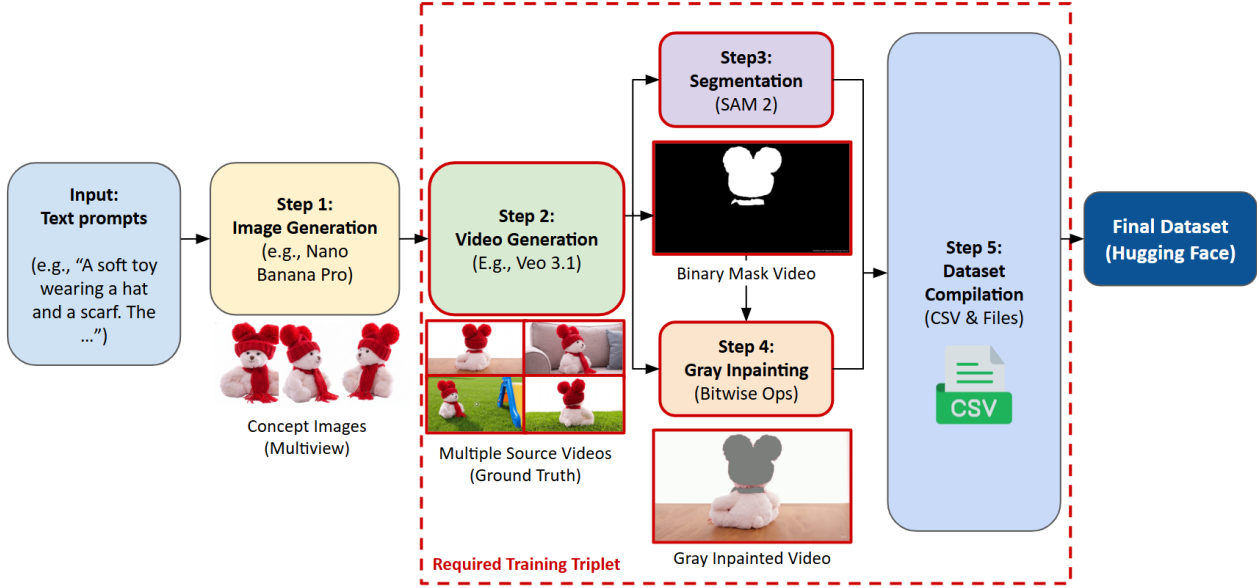


Figure 3. **Data Generation and Augmentation Pipeline.** We leverage Google Nano Banana Pro, Veo 3.1, and Meta SAM 2 to generate the triplet inputs for training.

2.2. Model training

We adopt **DiffSynth-Studio** as our codebase framework, which supports fine-tuning and inference across a variety of recent video and image generative methods. The repository also provides sample scripts demonstrating different generation workflows. We specifically leverage VACE [4] as the base framework for fine-grained concept learning since it is already capable of the video inpainting task and can take various forms of inputs, such as binary masks and prompts as conditioning. We aim to personalize the weights of VACE, given some source video and the mask video of the concept to learn (Fig. 1). We specifically finetune the VACE Control unit using LoRA [3] with a rank of 32 and keep the base Wan2.1 1.3B [11] DiT model frozen. We finetune for 5 epochs, with each epoch consisting of 100 iterations formed by repeating the dataset formed from the original source video and its generated augmentations. Inspired by [7], we further leverage a dual masked MSE formulation as given below.

$$\begin{aligned}
 L_{\text{masked}} &= \sum_t \|M_t \odot (\hat{x}_t - x_t)\|_2^2, \\
 L_{\text{unmasked}} &= \sum_t \|(1 - M_t) \odot (\hat{x}_t - x_t)\|_2^2, \\
 L_{\text{Dual}} &= \lambda L_{\text{masked}} + (1 - \lambda) L_{\text{unmasked}}.
 \end{aligned}$$

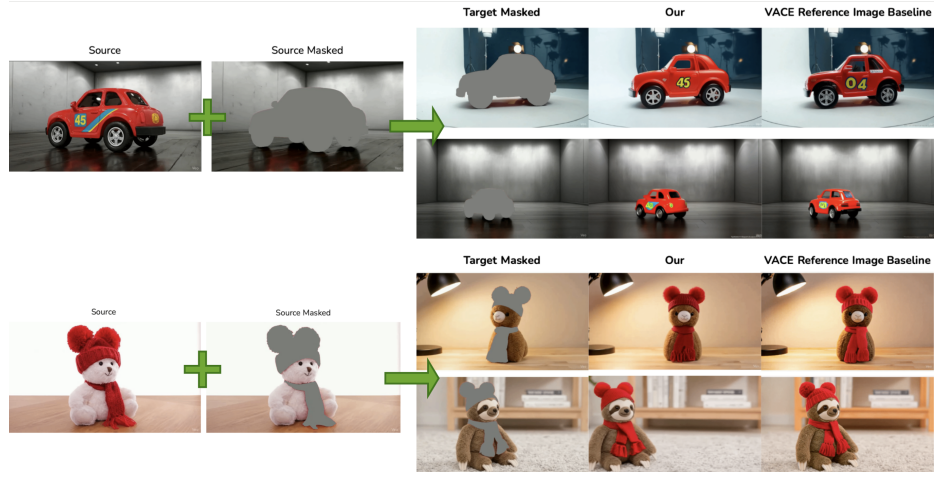
Here, M_t is a binary mask defined over the latent space of the model, where ones correspond to the target concept region and zeros correspond to the background. To enable more focus on the concept-relevant regions and less focus

on the background regions, we set $\lambda = 0.9$ in the above equation.

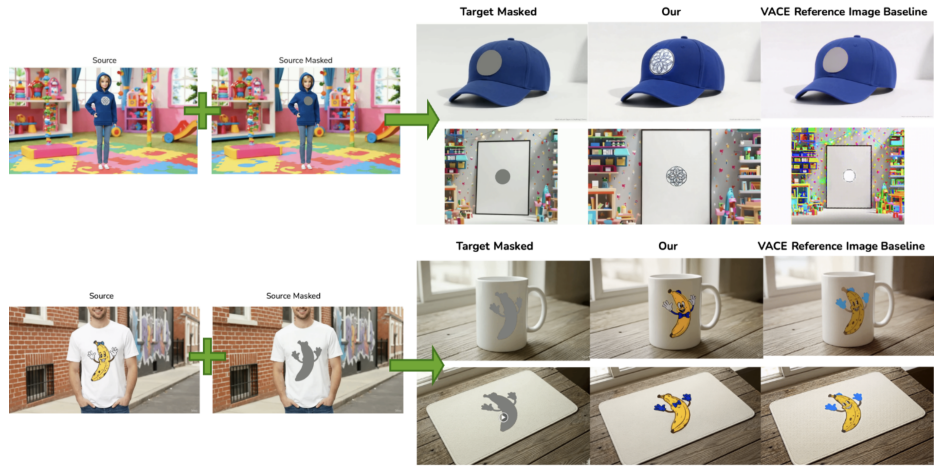
2.3. Cache Implementation

Upon inspection of the training and inference pipelines, we observed that a significant portion of the computation is repeatedly spent on encoding video inputs and text prompts into latent representations. To reduce this redundant overhead for each epoch, we introduce a caching mechanism that precomputes and stores the corresponding video and prompt tokens.

All experiments were conducted on an AWS `g6e.xlarge` instance equipped with an NVIDIA L40S GPU. The cached tokens were saved to disk and loaded directly during training. We also modified the data loader to optionally consume these cached representations instead of recomputing them on the fly. This design substantially improved the overall training efficiency, resulting in a 43% speedup: reducing the per-epoch runtime from 50 minutes to 35 minutes, without affecting model behavior.



(a) **top row.** Full object as a concept: exact replication is hard. **bottom row.** Simple cloth texture as a concept: simple textures are easy to replicate for both our method and the baseline.



(b) **top row.** Intricate logo as a concept: exact replication is hard, but our method captures the concept design very well. **bottom row.** Intricate logo as a concept: both our method and the baseline fail to exactly replicate the concept.



(c) **top row.** Furniture design as an abstract style concept: our method is able to capture the style of the source video. **bottom row.** Dynamic object (flag) as a concept: both our method and the baseline fail to exactly replicate the dynamic concept due to the complexity of the logo.

Figure 4. Experimental results across different concepts learned from the source and applied to different target objects.

3. Experiments

We conducted experiments on learning 6 different concepts with varying complexity (full object, simple pattern, complex logos, structural design, and a dynamic object). We compared our method with VACE’s [4] reference-image-based inpainting capability. Here, the reference image used for inpainting the target region was taken as the first frame of the source video, along with a prompt describing what was to be inpainted. Results and conclusions on different experiments:

- **Full object as a concept:** In Fig. 4 (a), the top row shows the results on learning a full object, in this case, the car with stickers, and transferring it to some other background. We can see that our method misses the details of the blue ribbon.
- **Simple pattern as a concept:** In Fig. 4 (a), the bottom row shows the results on learning a simple hat and a scarf. In this case, both our method and the baseline perform equally well.
- **Complex logo as a concept:** In Fig. 4 (b), the top and bottom rows show the results on learning complex logo designs as a concept. For the logo in the top row, our method captures the style really well and almost replicates the logo on the cap, whereas the baseline method completely fails to inpaint. In the case of the banana logo in the bottom row, both our method and the baseline add more details (color) on the hand regions, failing to replicate the exact details.
- **Abstract style as a concept:** In Fig. 4 (c), the top row shows the results on learning an abstract structural style as a concept and transferring it to some target object video. We can clearly see that our method excels in capturing the abstract style of the source objects, whereas the baseline method completely fails to capture abstract style concepts.
- **Dynamic object as a concept:** In Fig. 4 (c), the bottom row shows the results on learning a dynamic object, such as a flag, as a concept and transferring it to some target video. We can clearly see that dynamic concepts and exact replication are shortcomings for our method, and that our method mostly excels at abstract personalized style transfer.

We further summarize the design decisions that informed our final training and inference pipeline. Additional video results are provided in our project presentation slides, available at [VLR Final Presentation Slides](#).

3.1. Full MSE Loss vs. Masked Loss

The default VACE model training utilizes a standard full MSE loss across the entire spatial region of the latents, which tends to be dominated by background reconstruction errors rather than the target object. By implementing

a Masked MSE loss that calculates error strictly within the object’s binary mask, we force the model to focus on the fine-grained details of the concept.

3.2. Effect of Special Tokens: `ohwx` vs. `[V]`

We investigated the impact of the token identifier used to represent the target concept. Using the exact `[V]` as the special token interferes with the learning process since it already possesses a strong pre-trained semantic prior in the base model. As shown in Figure 5 (4th row), instead of generating the target logo, the model often yields a literal letter “V” or blends the concept with generic features. By switching to a rare token string identifier, specifically `ohwx`, we significantly mitigate this semantic conflict. The rare token acts as a neutral placeholder, allowing the LoRA layers to capture the unique visual identity of the new concept without fighting against the model’s existing linguistic associations.

3.3. Impact of Reference Image Conditioning during inference

While the visual texture of the concept is encoded within the finetuned weights, relying solely on text prompts during inference often proves insufficient for maintaining geometric stability. As observed in Figure 5 (5th row), inference without a reference image captures the correct pattern style but fails to replicate the exact style of the source video. We found that providing a reference image (e.g., the first frame of the source video) acts as a critical visual anchor. This hybrid approach drastically improves stability during inference, ensuring the generated video maintains better texture and structural fidelity to the original concept. This is equivalent to doing inference like the baseline method, but on our personalized weights, which capture the concept.

3.4. Necessity of the Text Prompt

Finally, we evaluated whether the finetuned model could generate the target concept based solely on the reference image and the learned weights, without an explicit text trigger. As shown in the last row of Figure 5, omitting the specific identifier `ohwx` from the prompt results in a failure to recall the target concept, and it even performs worse than the baseline which is inferred without finetuning. Instead of reproducing the target pattern, it hallucinates a generic logo-like object that roughly matches the shape but deviates in appearance—producing incorrect colors (e.g., blue instead of black). This confirms that the learned concept is tightly bound to the specific token embedding. The prompt is not merely a caption but a necessary trigger to activate the personalized LoRA parameters during the attention mechanism calculation.



Figure 5. **Qualitative ablation study on text tokens and reference image conditioning.** Compared to the Ground Truth (1st row), our method (2nd row) utilizing the `ohwx` token and a reference image achieves the best fidelity. The default token `[V]` (4th row) suffers from strong semantic priors, causing the model to generate a literal letter ‘V’ instead of the target logo. Without the reference image (5th row), the model captures the rough structure but fails to generate the exact same style. The baseline without finetuning (3rd row) lacks the sharp details of the specific pattern.

4. Conclusion & Future Works

In this work, we investigate LoRA-based concept learning within the VACE framework and observe that it is most effective for transferring high-level style and texture information. However, reproducing fine-grained and exact visual details remains challenging, indicating inherent limitations in the current adaptation mechanism.

We find that the method is highly sensitive to several hyperparameters, including prompt formulation, the choice of special tokens, and the LoRA rank. These factors significantly influence both concept fidelity and stability, suggesting that more systematic hyperparameter tuning and analysis are necessary.

Future work includes conducting more extensive ablation studies on where LoRA adapters are inserted. In particular, adapting DiT layers directly, rather than restricting modifications to the VACE control unit, may improve the learning of different types of concepts. Analysis of quantitative results should also be included. Additionally, exploring improved architectural designs and larger base models may help mitigate the observed limitations and further enhance concept transfer quality.

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